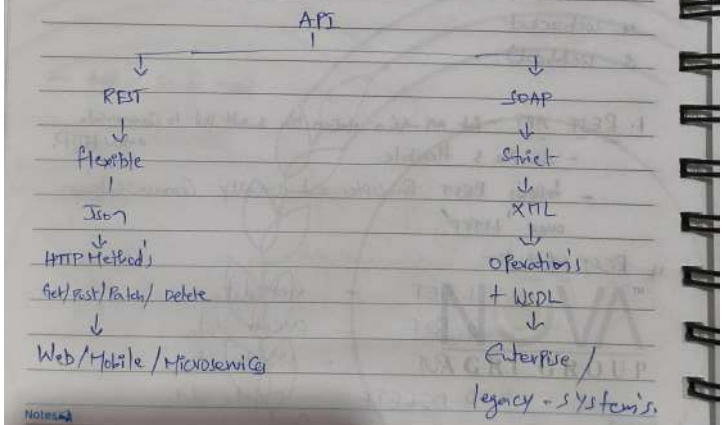
2 SOAP API (Simple Object Access Protocol)

- Structured & Strict.

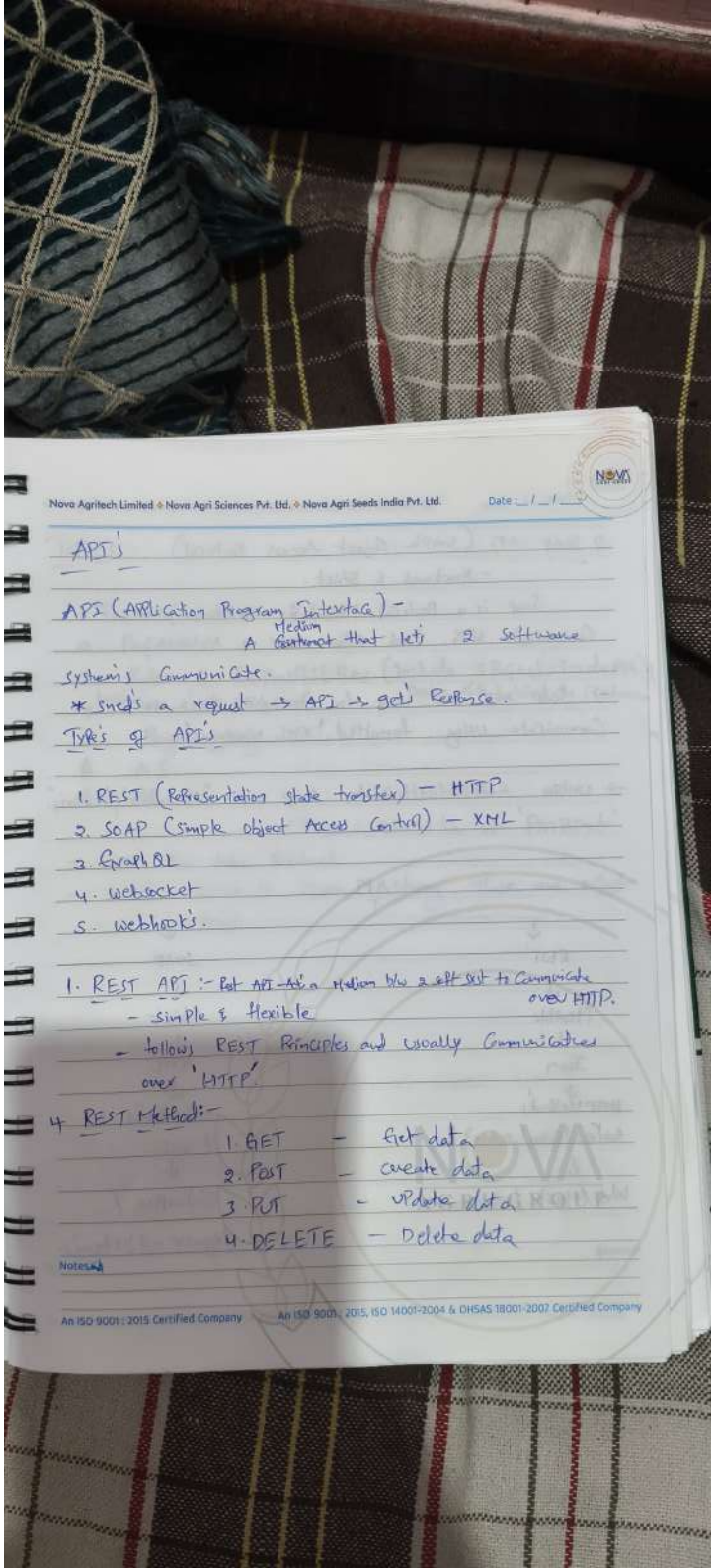
Soap is a Protocol used by Applications to Communicate with each other over a Network.

→ Soap API is a way for two Application's to Communicate using formatted 'XML Messages'.

→ Unlike Get/Put/Post/Update, Soap uses "Operation's/Action's."



Notes:



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API's

API (Application Program Interface) -
A Medium that let's 2 Software systems Communicate.

* sends a request → API → gets Response.

Type's of API's

1. REST (Representation state transfer) – HTTP
2. SOAP (Simple object Access Control) – XML
3. GraphQL
4. Websocket
5. Webhook's.

1. REST API :- ~~Set API - Act a Medium b/w 2 soft set to Communicate~~ over HTTP.
   - simple & Flexible
   - follows REST Principles and usually Communicates over 'HTTP'

4 REST Methods:-
   1. GET – Get data
   2. POST – create data
   3. PUT – Update data
   4. DELETE – Delete data

Notes:

2. Project's (learn what's going on behind the Picture)

1. Telugu Voice AI Assistant :-

User speaks (Query Raises)
↓
Speech to text (Fast API : Post/stt)
(Sarvam STT)
↓
Translate Query (Telugu text to English)
↓
Retrive from ChromaDB
↓
LLM Generated Answer
↓
Translate Answer (English back to telugu)
↓
Text to speech (Fast API : Post tts)
↓
User's hears Answer.

→ I have used Azure for Deployment.

Transformer Layers - 8, one layer Process, Next layer Process updated Attention + feed forward. Refinement

Attention heads - 8

Context length - 128 - Max no of tokens model can process together by one input.

AI/ML

1. AI :-

Artificial Intelligence is a broader field of "building systems that can perform tasks that normally require some form of Human Intelligence".

2. AI - ML - Deep learning.

AI - The broadest concept

AI
└ systems performing Intellegent tasks.

Ex:- chatgpt, self driving car, Siri.

ML -

"A subset of AI where systems learn Patterns from data" rather than relying entirely on explicitly Programmed Rules

AI
└ ML

Ex:- fraud detection, stock Price Prediction, House Price Prediction.

Deep learning -

"A subset of ML that Primarly uses Multilayer Neural Networks to learn Complex Patterns".

AI
└ ML
└ Deep learning.

Ex:- Give 1000 images to DL it learns. eyes, nose, facial shape, soon Recognize New Image

EX:- self driving car, Image Creation, Voice Recognization.

Notes:

Batch Size :-

Batch size is the Number of training Examples Processed before an optimizer update.

Dataset - 10,000 samples
Batch Size - 100

Small batch	Large batch
less Memory	More Memory
Can be slower Per epoch	Can be Computationally Efficient on suitable hardware.

How Everything Connects :-

Training data
↓
Neural Network
↓
forward Propagation
↓
Prediction
↓
loss function
↓
loss
↓
Back Propagation
↓
Gradients
↓
Optimizer → Updates weights / Bias → Next Batch → Repeat batches

Repeat Epochs.
↑
Next Epoch



Deep Learning

Epoch :-

An epoch Means one Complete Pass through the entire training dataset..

Ex:-

Training Samples - 1000

- if the Model sees all 1000 Samples once - 1epoch.
- if it sees all 1000 Samples 5 times - 5 epochs.

Batch :-

A batch is a subset of training data Processed together in one training step/update.

Ex:- Dataset = 1000 Samples

Batch Size = 100

Batch-1 - 1000 100

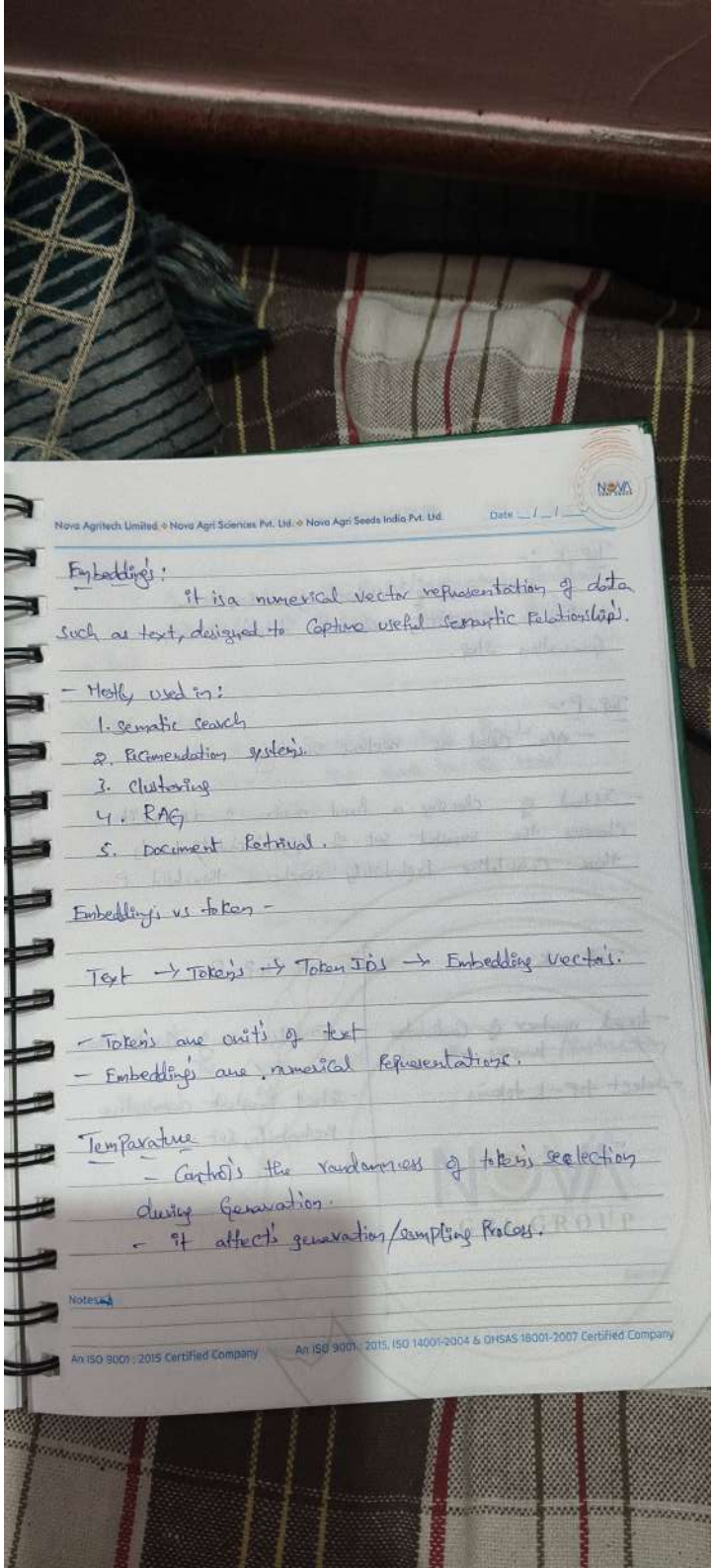
Batch-2 → 200

⋮

Batch 10 → 1000

Therefore 1 epoch = 10 batches/steps.

Notes:



Embeddings:

it is a numerical vector representation of data such as text, designed to capture useful semantic relationships.

- Mostly used in:
  1. Semantic search
  2. Recommendation systems
  3. Clustering
  4. RAG
  5. Document Retrival.

Embeddings vs token -

Text → Tokens → Token IDs → Embedding vectors.

- Token's are unit's of text
- Embeddings are numerical representations.

Temparature

- Control's the randomness of token's selection during Generation.
- It affect's generation/sampling Process.

Notes:



Tokenization

It is a Process of Converting text into tokens that the Model Can Process.

- The tokens are Converted into ID's.

Token's → Token ID's.

Ex:-

AI → 245 1

is → 318

amazing → 4567.

- These Numbers then used to retrive / Construct Embeddings.

Encoder	Decoder
- The Encoder builds Rich Representations of Input - understanding / Classification	- It is designed for generating output tokens. - Text Generation.
Ex:- Text Classification Sentiment Analysis.	Ex:- GPT style Models.

Tokens

A token is a unit of text Processed by a LLM.

- Token can represent whole words, Part of words, Punctuation, or other text units depending upon tokenizer.
- Punctuation and spaces can also affect tokenization.
- LLM's Generally Process Tokens, Not raw words.

Why tokens matter?

Tokens affect :-

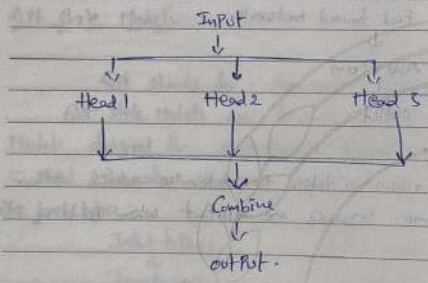
- Context window usage
- API Cost's
- Processing time
- Max Prompt Length
- Max output Length.

Notes:

Multi-Head Attention.

Instead of performing only one attention calculation, Transformers use Multiple attention heads - Each head can learn diff types of relationship's.

Head-1 :- grammatical related
Head-2 :- Nearby word "
Head-3 :- Coogroye "
Head-4 :- Semantic "



def:-

Multihead attention Performs attention in Multiple learned representation subspaces in Parallel, Allowing the Model to Capture diff relationships b/w tokens.

Notes:

Gen AI / LLM

LLM (Large language Model)

An LLM is a Neural Network "trained on a very large Amount of text to learn Patterns in language and generate Human like text".

Ex:- models from families such as Gpt, Claude, Gemini

Gpt Style Models:

Gpt stands for - Generative Pretrained transformer.

Gpt style models are Generally decoder only Transformer Models designed for Auto Agressive ~~Regression~~ generation.

- Model Generates Answer token by token.

The basic idea :-

Input tokens
↓
Transformer
↓
Predict next token
↓
Add token to Context
↓
Predict next token
↓
Repeat.

Ex:- Gpt3, Gpt4, Gpt5, Llama, Qwen.

Notes

Factais Can Include:

1. Missing Information
2. Ambigeous Question's
3. Poor Prompt's
4. outdated knowledge
5. Insufficient context
6. Long/Complex task's
7. Retrival failures.

How Can you reduce hallucinations?

1. RAG
2. Better Prompt's
3. Structured outputs
4. Validation against trusted sources.
5. Lower Randomness when Appropriate.
6. Citations / Source grounding.

Notes

intended instructions.

Hallucination

LLM hallucinates when it generates information that appears plausible but is incorrect, unsupported (or) fabricated.

Ex:- Who invented XYZ technology? - The Model Confidently Provides a Person name who never invented it.

Why does hallucination happen?

LLMs generate outputs based on learned patterns and the information/context Available to them. They dont Automatically verify every statement against a trusted database for reality.

Notes:

Prompt Engineering.

Zero shot Prompting - None

ZSP means Asking the Model to Perform a task without Providing an Example

one-shot Prompting - one

O-SP Provides one Example before asking the Model to Perform the task.

Few-shot Prompting - Several

It Provides Multiple Examples.

Chain of Thought Concept

It Refers to Step by step Reasoning used to solve a Complex Problem.

Role Prompting

R.P tells the Model what Role (or) Prospective to adopt.

Ex:- You are an Experienced Aws Cloud Engineer.
Explain Aws IAM ?

Notes:

*How do you know your RAG system is working well?

I evaluate a RAG system using different Metrics, such as

1. Retrival Accuracy - Precision@k, Recall@k
2. faithfullness
3. Answer Relevance
4. Context Precision.
5. Monitor latency, Response quality, hallucination rate, in Production.

Notes:



Chunking

Chunking means splitting a large document into smaller pieces of text.

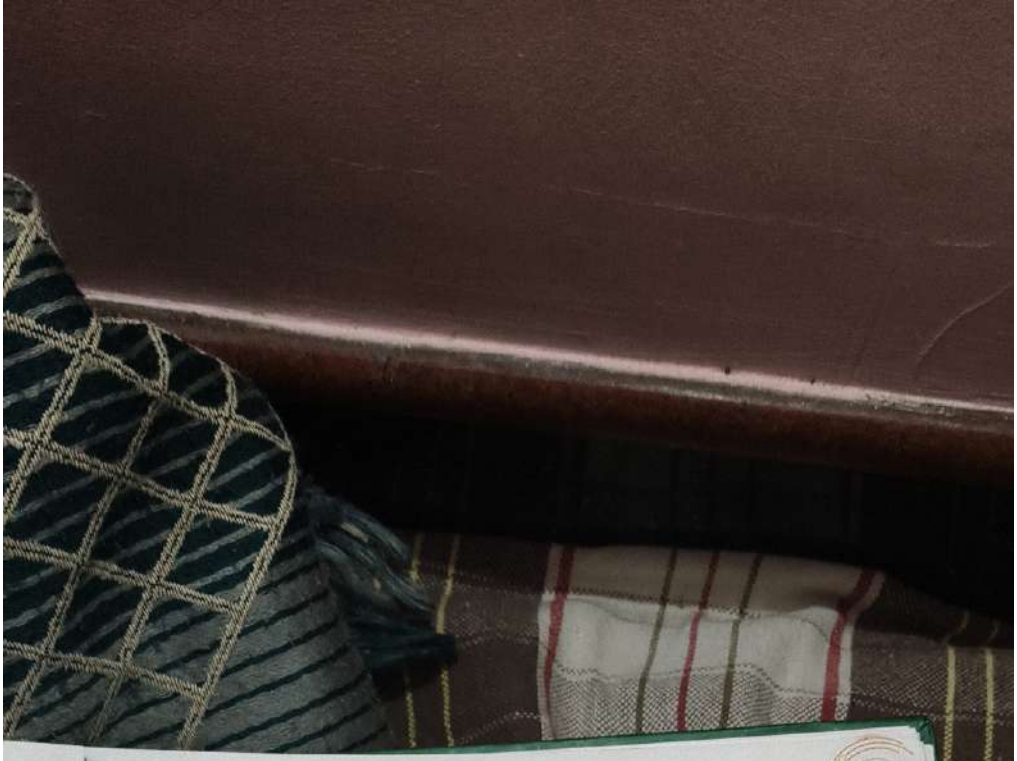
Embeddings

An Embedding is a numerical vector representing text (or) other information.

Retrival

It is a process of finding relevant information from the indexed K.B

Notes



2. Why we need RAG?

1. Reduce hallucination
2. latest information
3. Cost Effective
4. Better Accuracy.

3. RAG

1. Requires a retrival pipeline
2. Good for Private changing knowledge
3. Update knowledge by Updating sources

Fine tuning.

1. Requires a training dataset and workflow
2. Good for Adapting style, task behaviour, domain Patterns.
3. Requires another training for fine tuning Process

4. Document ingestion

it is a Process of bringing your data into RAG system and Preparing it for Retrival.

Sources Include :-

1. Pdf files
2. word document
3. HTML Page, Text file
4. K.B, Databases.

Notes



RAG (Retrival - Augmented Genaration).

RAG

it is an Approach that Combines information retrival with an LLM to genarate answers.

Retrival - find Relevant information from document's / K.B
Augmentation - add the Retrived information to LLM Prompt's
Genaration - let the LLM Produce an Answer using the question and Retrived Context.

User Asks Question
↓
Embedding
↓
vector search
↓
Retrive Context
↓
LLM Genarated Answer
↓
final Answer.

User Asks Question
↓
Retrive Relevant document's
↓
Add Context to Prompt
↓
LLM Genarated Answer.

Notes

Encoder - Understands the Given text.
A Neural Network Process input text and understands Relationship b/w token's.

Decoder - Generates the Next token.
A Transformer Architecture that generates text, one token at a time by Predicting the Next token using the Previous token as Context.

LLM (Large language Model)

It is an AI Model that's trained on large Amount of text data to learn language Pattren's, Relationship b/w token's and Predict Next token in sequence.

In simple - learn's Pattren's from text, and use those Pattren's to understand & Generate language.

Notes



Masking'

— it is a technique that Prevents Transformer from Accessing Certain info which it shouldn't.

Masking - Hiding selected word's/token's from the Attention Mechanism.

Ex!- Imagine you are writing a Exam, A Question:-
'The Capital of India ——'
You need to fill the blank, But Imagine teacher shows you Answer before you Even Attempt.

→ This is where 'Masking' Comes.

Tokenization

it is a Process of Converting text into smaller token's Called token's.

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Notes

Query - what am i looking for? (Question)
key - How relevant is this item? (Matching)
value - what info do i get? (Info).

Multihead Attention

it allows transformer to perform multiple attention operations in parallel, using different learned ~~vector~~ Projections. allowing model to learn diff Relationships b/w tokens.

self Attention - one Attention Operation
Multihead Attention - Multiple Attention operations Running in Parallel.

Notes:



② Librarian Check's label's & titles of diff books

Book	Key
Book-1	Python
2	AI
3	Coding

Librarian Compares your search request with Label's.
This is the Purpose of Key.

Key - What does each item offer?
↳ Info used to determine book Matches search.

③ Librarian finds and gives to you.
You open book and get Actual information.
This info is Value.

Value - What Info do i get?
Actual Content of selected book.

I love AI - Transformer Recieve these token's.

Input - [I] [Love] [Artificial] [Intelligence] - tokens

↓ Numerical Representation (Each token has vector's)

↓

[Q] [K] [V]

*Every token has it's own Q, K, V's.

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Self Attention.

It is a Mechanism that allows transformer to understand Relationship b/w diffrent words (token) in a sentence.

Ex :- bank → [bank
River bank.

Query, Key and Value (Q/K/V)

Query - what am I looking for ?
Key - what does each item offer ?
Value - what Infomation do i get ?

Imagine you are in a Library, You want book about AI. There are 1000 book's.

① You ask Librarian what am i searching for

I want a book about AI → This is our Query.

Query - what am I looking for - A book About AI.

Notes

Gen AI / ML

Transformer
- it is a type of deep learning Architecture
- it helps Computer to understand and Generate Human Language.

Architecture :-

Input text
↓
Tokenization (split text into tokens)
↓
Embeddings (Convert's tokens into vectors)
↓
Positional Encoding (Add word order information)
↓
Transformer layers (understand's context)
(Self Attention, FF Network)
↓
output Predictions
(Generate or classify text)

Notes

ROS (Robot operating system)

it is a Robotic software framework that Provides tools and Communication mechanism to build and Control Robots.

Imagine you are building a Robot, it has

- Camera
- Sensors
- Arms
- Microphone

→ All these Need to Communicate with Each other Right,
→ ROS helps to Connect & ~~Communicate~~ & Coordinates these Components.

NLP - understand human language
LLM - understands and Generate language using trained Model.
VLM - understands image and text & Generate text Responses.
VLA - Connects vision, Language and Physical Actions.
ROS - Connect & Coordinate Robot Software & Hardware

Notes:

→ How does VLM read an Image?

Image Input

↓

Image Processing

1. Resize image 2. Normalize Pixel values 3. Prepare Image for Model.

↓

Divide into Patches

(Image gets divided into smaller Patches)

↓

Vision Encoder

A Neural Network (often a vision Transformer) analyze Patches

↓

Image Embeddings

Converts visual Info to understandable Answer Question.

↓

Language Model.

Notes



→ How does LLM Read text?

Text Input
"I love AI"
↓
Tokenization
["I", "love", "AI"]
↓
Token ID's
[40, 2031, 1589]
↓
Embeddings
Numerical vectors Representing tokens
↓
Transformer Process Embeddings.

Notes

VLM	VLA
Vision Language Model	Vision Language Action
- Understands image & ~~Action~~ language	- Connects vision and langue to Action.
- Describe and Answer Questions	- Helps Control Physical Action
- Image Question Answering	- Robotic Manipulation.
If you show a robot A Red bottle.	
→ I Can See a Red bottle on table.	→ See's Red bottle, understand to Pick, Generate Actions and Pick it up.
⇒ only says what it see NO Action.	
VLM = See's and understand	VLA = See's understand & Act.

How Computer Process Image?

Image Input → Image decoding → Pixel Representation ↓

Processed Image. ← Image Processing ← Images as an array

Notes

LLM (Large Language Model)

it is a AI Model where it trained on Huge amount of text data to understand and Generate Human Language.

VLA (vision Language Action)

it helps AI-Powered robot's to understand their Sourounding's, follow instruction and Perform Physical Action's. (sees - understad - Act)

Vision + Language + Action.

You tell robot to Pick Red bottle.

The VLA':
- Sees the bottle Using Camera
- understad's instruction,
- Determine Required movement
- Generates Action.

Notes:

NLP (Natural language Processing)

it is a branch of AI that helps Computer understand, Process and work with Human language. (deals text only).

- it helps Computer to understand human language.

* it helps Computer Process those human lauage requests.

for ex: you tell Computer to book Ticket to Hyd.

- NLP helps Computer to understand
  - what you are asking
  - what Action you want
  - what info it needs.

* "Helps Computers understand and work with human language".

Ex) - Google translate, voice Assistant.

NLP helps Computer - understand - Process - work with H.L.

Notes

VLM (vision language Model)

it is a AI Model that can understand visual Information and connect it with Human language. (Sees - understad).

→ If i gave a image and ask a question :-
- it Analyze, Read's, (or) Process the Image and Gives Info in our language.

Where VLM's are used ?

1. Self driving Cars
2. Robotics and Automation.
3. HealthCare.

→ VLM is a Multimodel AI System, it Can Process and Respond back Image (vision) and text (language).

differences

	Input	output
LLM	Text	text
VLM	Image + Text	text.

Notes:

Image → vision Encoder → Projector/Creator → Multimodel LLM → Answer

FPS (frames Per second)

- how many frames displayed in one (1) second.

- Video has 30FPS, That Means

30FPS
~~(Second~~ - 30 Frames Per Second
60 FPS - 60 frames Per second.
1 FPS - 1 frame Per second.
120 FPS - 120 frames Per second.

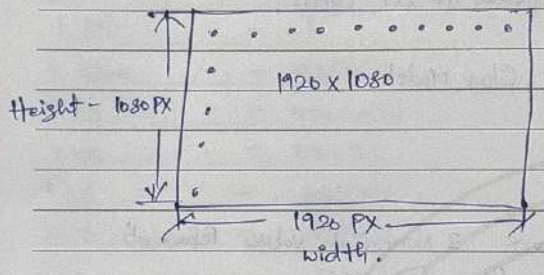
→ How video stored inside a Computer?

Video
↓
frames
↓
Pixels
↓
RGB Values

Notes

Image Resolution

No of Pixels Present in an image.
Expressed in 'width x height'.
- it tells how many pixels image can contain.
- More Pixels allows Image to Preserve More fine details.



1 MP = 1,000,000 Pixels.

Video

Collection of images Played Continously.

Frame

A Single Individual image inside a video.

Image - 1 Photograph, Frame - one image inside video
Pixel - Image Made of tiny Million squares
~~Fps - frames displayed per second~~
Video - Collection of frames Played Continously.

Notes
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Image

It is a visual Representation of something that we see in Real world.

→ An Image is Made up of Millions of tiny squares called Pixels.

Pixel – Picture Element.

How Computer Represent (or see) Colours?

→ it uses RGB Colour Model.

R - Red
G - Green
B - Blue.

→ Every Pixel Can have 3 Numerical values Representing Red, Green, Blue, Ranges 0 to 255.

Red – RGB (255, 0, 0)
Green – RGB (0, 255, 0)
Blue – RGB (0, 0, 255)
White – RGB (255, 255, 255)
Black – RGB (0, 0, 0)
Yellow – RGB (255, 255, 0)
Magneta – RGB (255, 0, 255)

Notes

A single binary ~~bit~~ digit is called Bit.

Ex:-

1 = 1 bit

1010 = 4 bits

11001010 = 8 bits.

unit		size
1 Bit	–	1
1 Byte	–	8 bits
1 KB	–	1024 bytes
1 MB	–	1024 KB
1 GB	–	1024 MB
1 TB	–	1024 GB

Notes

This is where GPU Comes UP!

GPU (Graphic's Processing unit)

It was designed to Perform Many Calculations in Parallel (ow) to Perform huge task's.

- it Can Process set of images/video at High Resolution,
  it Can Read the images in a Higher Rate.
- So, we Can simply play Call of Duty, Pubg with Ease.

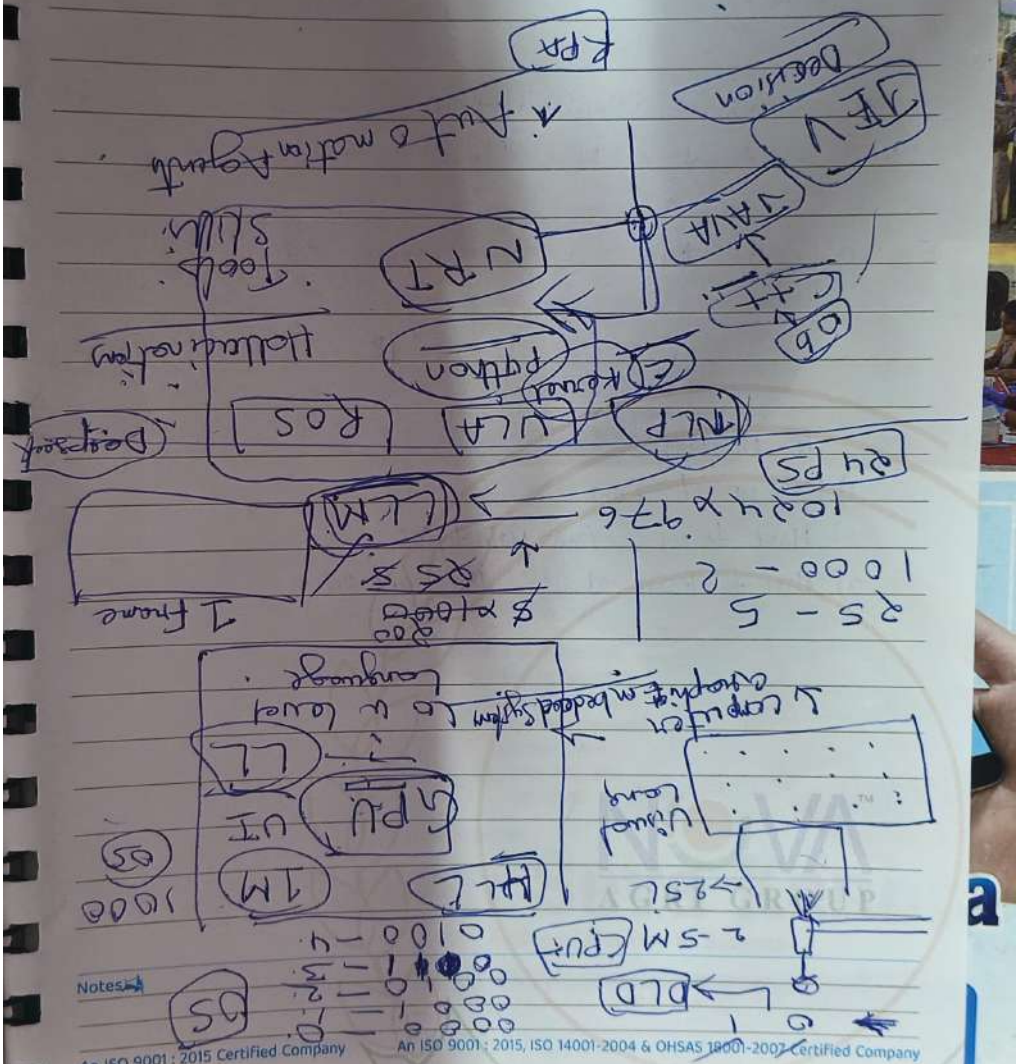
Binary Number's

It is a Number system that uses only two Number's 0 and 1.

Binary Mean's two.

0 - Off / low voltage
1 - ON / High voltage.

Notes





~~How we are using CPU in our Project?~~

→ A Computer Doesn't understand, Photo's, video's, laungages, sound's. 'it only understands numerical information'.

for suppose if i gave three things to Computer it Process them into:

1. Image - Numerical Pixel values
2. Sentance - text to token, they token to Numerical Repre
3. Audio - Numerical Audio Signals.

Now let's see how Computer Hardware Perform these Actions.

1. CPU (Central Processing unit)

it is Responsible for Executing instruction's and Performing General Purpose Computation's.

- Generally used for small type of work,
  Ex:- it Can suppot small games that we Can Play like Roaddash, Card's,
- Cannot Process high Resolution's i.e. Cannot Process Set of images/video for high Resolution.